

## 8. Relevance of Training to Academic Curriculum

This training practically applied several theoretical subjects from the GTU Computer Engineering curriculum, notably: Database Management Systems (ACID properties and schema design in SQLite), Mobile Application Development (Activities, Services, and Layouts), and Computer Networks (TCP/IP, HTTP protocols, and WebSockets).

## 9. Overall Experience (Self-Reflection)

The internship at Netsol IT Solutions Pvt Ltd was a highly demanding and rewarding experience. Transitioning from basic Android apps to building a complex, distributed Server-Client architecture taught me how to think like a systems engineer. It bridged the gap between academic theory and real-world enterprise constraints (like OS security policies and memory management).

## 10. Suggestions (if any)

Future iterations of this project could benefit from implementing a Master Supervisory Node that aggregates data from multiple servers, as well as adding a secure authentication layer to the web dashboard to prevent unauthorized local network access.

### Signatures

Student Signature: \_\_\_\_\_ Date: \_\_\_\_\_

Industry Mentor Signature & Seal:



Faculty Coordinator Signature: \_\_\_\_\_

I observed the strict threading rules enforced by the Android OS. Specifically, network operations (like listening for WebSocket messages) execute on background threads, and attempting to directly update the UI from these threads causes instant application crashes. This reinforced the necessity of using `runOnUiThread()` for all visual updates.

## 5. Challenges Faced:

- **Challenge:** Formatting complex relational database data for efficient network transport without causing lag.
- **Solution:** Instead of sending multiple individual messages, I utilized the Gson library to serialize Java `List<Room>` objects into lightweight JSON arrays, packaging the entire database state into a single, highly efficient `REFRESH_ALL` payload.

## 6. Technologies & Tools Explored:

- Android Studio (Java & XML)
- NanoHTTPD (Embedded Server)
- Java-WebSocket library
- DB Browser for SQLite
- Postman (for local API testing)

## 7. Interaction with Mentor / Team:

Engaged in deep architectural discussions with my industry mentor regarding the SSOT model. We debated the pros and cons of cloud vs. local processing for medical hardware, ultimately deciding that a localized server prevents catastrophic failures during external internet outages.

## 8. Key Milestones Achieved:

- Successfully hosted a live web page directly from a mobile device (port 8080).
- Established bidirectional WebSocket communication between two Android devices on the same Wi-Fi network.

## 9. Next Steps / Action Items for Week 2:

- Integrate a custom audio engine for medical alarms (Code Blue).
- Bypass aggressive Android battery optimization techniques that kill background servers.
- Add priority sorting and grouping functionalities.

Sign of Head :



Seal of Organization:



Supervisor / In-charge:



- **Challenge:** Xiaomi MIUI and other strict Android ROMs silently blocked the Server app from connecting to its own WebSocket via the 127.0.0.1 loopback IP address, breaking the Server's UI.
- **Solution:** I decoupled the local screen from the network stack. Instead of forcing local UI updates over a socket, I triggered a native Android Intent broadcast precisely when the HTTP server fired the external WebSocket update.
- **Technologies & Tools Explored:**
  - Android Services and Manifest Permissions
  - Chrome Developer Tools (Network Tab)
  - Advanced SQLite queries and database migrations
  - JavaScript native printing capabilities
- **Interaction with Mentor / Team:**

Collaborated closely with my mentor to debug the audio engine's "Amnesia Loop" (where alarms restarted infinitely). Under their guidance, I decoupled the audio logic from the database and implemented a local tracking flag to accurately count audio repeats based on the administrator's settings.
- **Key Milestones Achieved:**
  - Stabilized the background service, achieving 24/7 uptime even when the Android device is locked.
  - Converted the dashboard into a fully headless REST API configuration panel.
  - Successfully generated the final Release APK (v1.1.0) and completed the internship project.
- **Overall Learning:**

Week 2 fundamentally changed how I view mobile application development. Moving beyond simple UI screens, I learned how to treat an Android device as a robust, localized networking hub. Dealing with OS-level memory management, network security policies, and HTTP caching provided me with enterprise-level debugging skills that cannot be replicated in standard academic projects.

Sign of Head :



Seal of Organization:



Supervisor / In-charge:



## Annexure-4: SKILL BASED TRAINING COMPLETION CERTIFICATE



**Netsol IT Solutions Pvt. Ltd.**

Mindset Of Positive Expectancy  
(AN ISO 9001 : 2015 Certified Company)



GeM  
Government  
e Marketplace



**NSIC**

### TRAINING EXPERIENCE CERTIFICATE

**Date: 09<sup>th</sup> July 2026**

This is to inform whomsoever it may concern and certify that **Mr. Pratham Solanki, Trainee of Software Engineering in Mobile Department** from **15<sup>th</sup> June 2026 to 30<sup>th</sup> June 2026**, has completed his training tenure as per company's record.

During his training period, we found **Mr. Pratham Solanki** to be Professional, Knowledgeable, and Result Oriented with a theoretical and practical understanding of work requirements.

He has a friendly, outgoing personality, a very good sense of humor, and works well as an individual or member of a team as required by the management. Overall, **Mr. Pratham Solanki** performed his duties and responsibilities cheerfully with attention to detail all the time. With his enthusiasm to work, learn, and progress, I am certain that he would make a great employee at any enterprise.

For, **Netsol IT Solutions Pvt. Ltd.**



Authorized Signatory

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